CHEMISTRY TERMS

1. MATTER:

Anything that has mass and occupies space is called matter.

Example:

Air, water, stone, wood, etc.

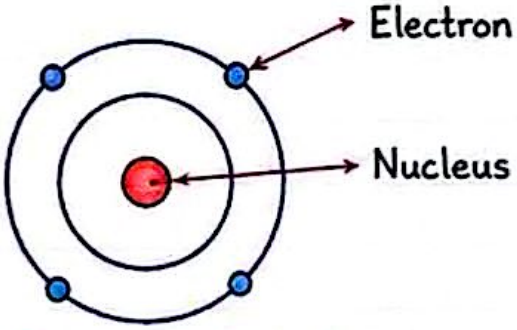


2. ATOM:

The smallest particle of an element that can take part in a chemical reaction.

Example:

Hydrogen atom, Oxygen atom.



Structure of an Atom

3. MOLECULE:

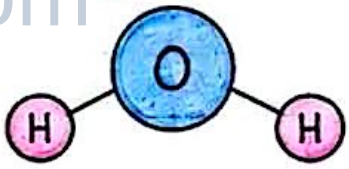
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Two or more atoms chemically combined together are called molecules.

Example:

H_2O (Water), O_2 (Oxygen), CO_2 (Carbon dioxide).

H_2O (Water)



CO_2 (Carbon dioxide)



4. ELEMENT:

A pure substance made up of only one type of atom. It cannot be broken down into simpler substances by chemical means.

Example:

Gold (Au), Iron (Fe), Oxygen (O_2).

Some Elements

- Hydrogen (H)
- Oxygen (O)
- Carbon (C)
- Iron (Fe)
- Gold (Au)

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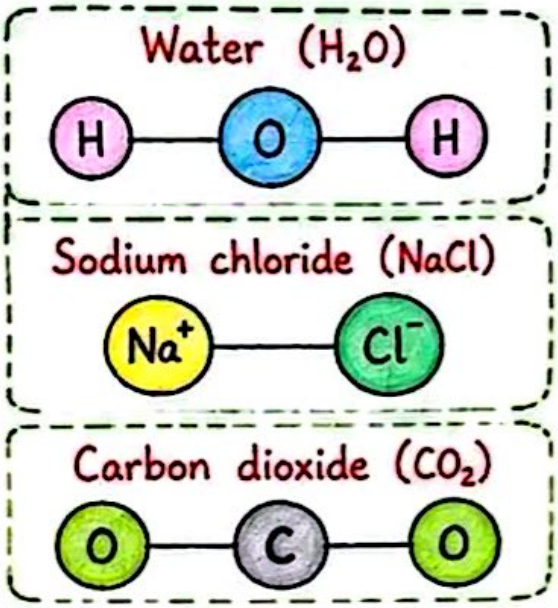
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5. Compound:

A substance formed when two or more different elements are chemically combined in a fixed ratio.

Example:

Water (H_2O), Sodium chloride (NaCl), Carbon dioxide (CO_2).

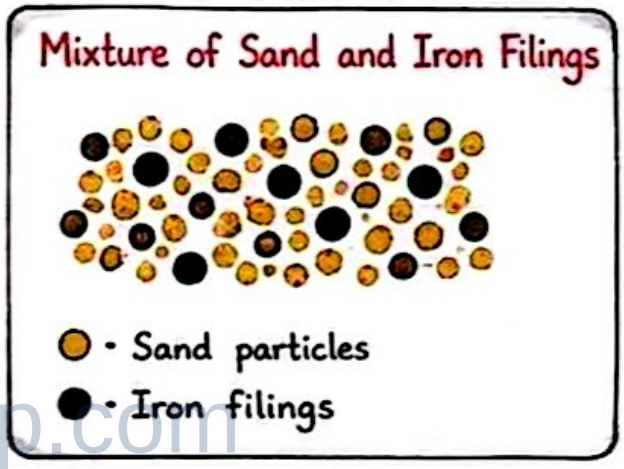


6. Mixture:

Two or more substances mixed together physically in any proportion. The components retain their individual properties and can be separated by physical means.

Example:

Air, salt in water, sand and iron filings.



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7. Pure Substance:

A substance that has a fixed composition and definite properties throughout. It is either an element or a compound.

Example:

Distilled water, Oxygen gas.



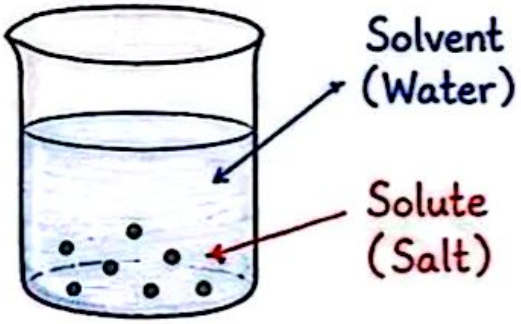
Pure distilled water (H_2O)

8. Solution:

A homogeneous mixture in which one substance (solute) is dissolved in another (solvent).

Example:

Salt in water, Sugar in water.



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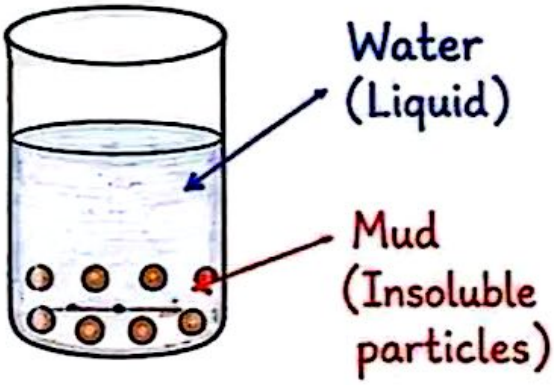
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9. Suspension:

A heterogeneous mixture in which insoluble particles are large enough to settle down if left undisturbed.

Example:

Mud in water, Chalk in water.



10. Colloid:

A mixture in which particles are intermediate in size between those in solutions and suspensions. They do not settle down.

Example:

Milk, Fog, Smoke, Paint.



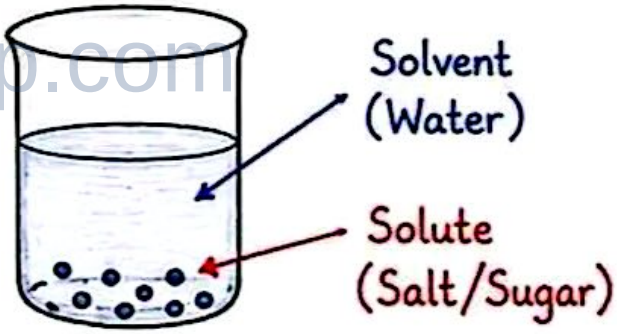
11. Solute:

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The substance that gets dissolved in a solvent to form a solution.

Example:

Salt in water, Sugar in water.



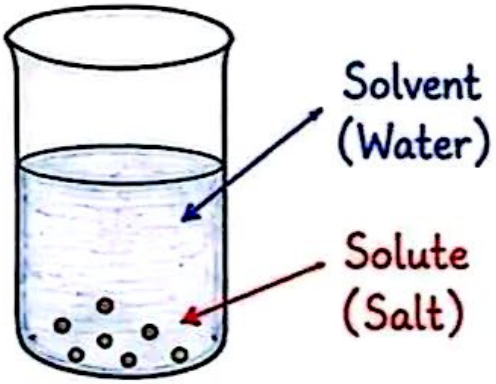
12. Solvent:

The substance that dissolves a solute to form a solution. It is usually present in larger amount.

Example:

Water in salt solution,

Ethanol in iodine solution.



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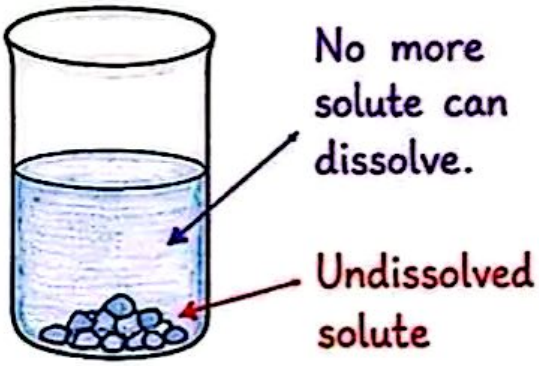
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13. Saturated Solution:

A solution that contains the maximum amount of solute that can dissolve in a solvent at a given temperature.

Example:

Saturated salt solution at 25°C.

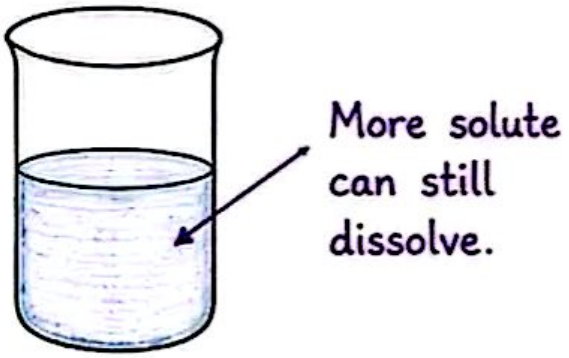


14. Unsaturated Solution:

A solution that contains less solute than the maximum amount that can dissolve at a given temperature.

Example:

Salt water before more salt is added.



15. Supersaturated Solution:

A solution that contains more solute than the normal maximum amount that can dissolve at a given temperature.

Example:

Prepared by cooling a hot saturated solution slowly.



★ These solutions are unstable. Extra solute may separate out easily.

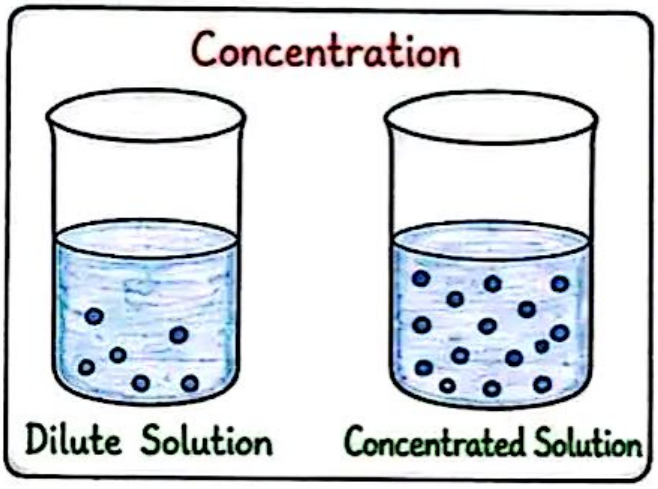
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16. Concentration of Solution:

It tells how much solute is present in a given amount of solution or solvent.

Types:

- Dilute Solution
- Concentrated Solution
- Molarity



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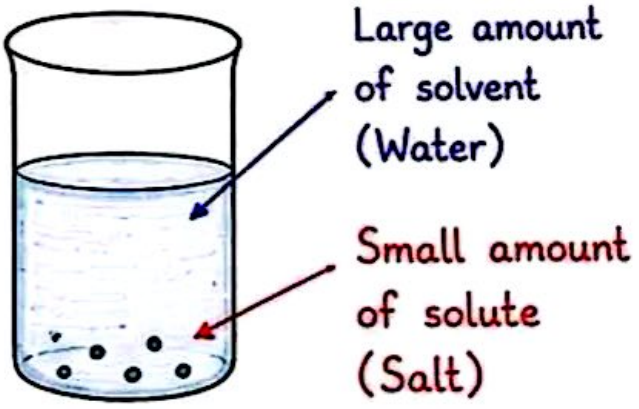
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17. Dilute Solution:

A solution that contains a small amount of solute in a large amount of solvent.

Example:

One spoon of salt in a full glass of water.

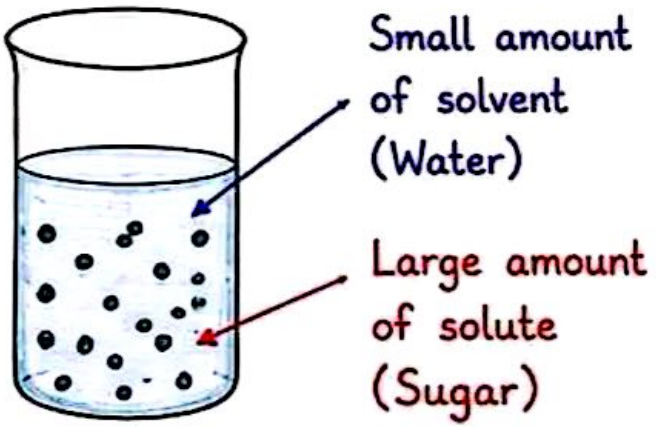


18. Concentrated Solution:

A solution that contains a large amount of solute in a small amount of solvent.

Example:

Many spoons of sugar in a small glass of water.



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19. Molarity:

It is the number of moles of solute present in 1 litre of solution.

Formula:

$$\text{Molarity (M)} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}}$$

Mole:

A mole is the amount of substance that contains 6.022×10^{23} particles (Avogadro number).

Example:

If 1 mole of NaCl is dissolved in 1 litre of water, then the molarity of the solution is 1 M.

Common Molar Solutions:

- 1 M NaCl solution
- 1 M HCl solution
- 0.1 M NaOH solution

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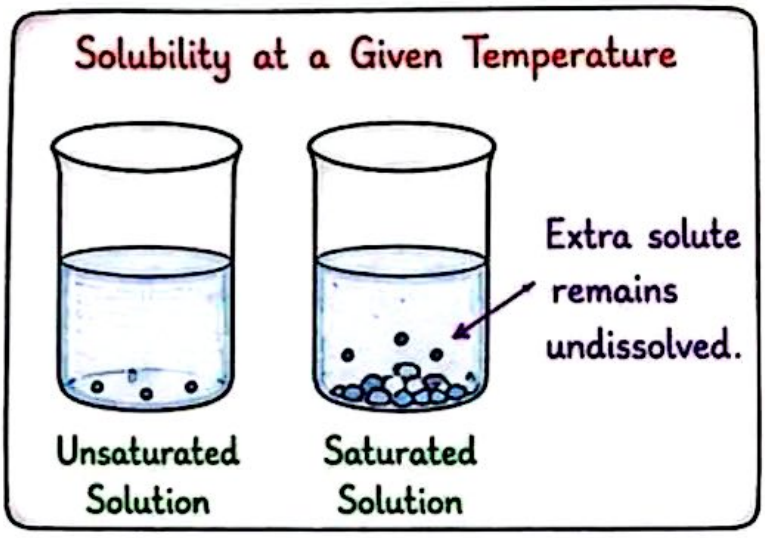
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20. Solubility:

The maximum amount of solute that can be dissolved in a given amount of solvent at a specific temperature.

Example:

At 20°C, about 36 g of NaCl can be dissolved in 100 g of water.

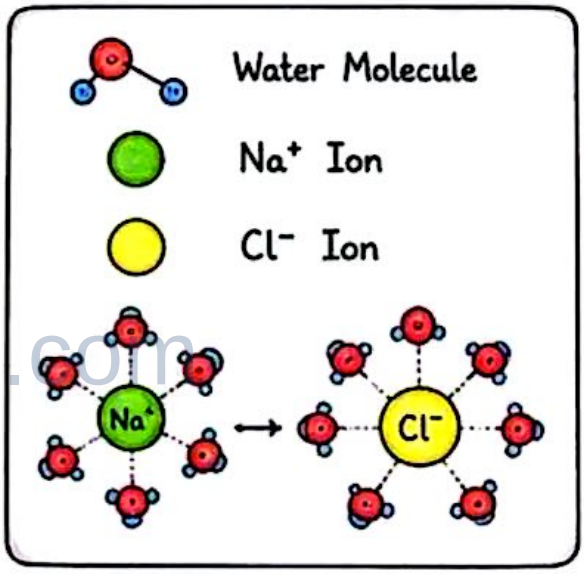


21. Solubility and Solute-Solvent Interaction:

Solubility depends on how strongly solute particles attract solvent particles.
If the attraction is strong, solubility is high.

Examples:

- Iodine dissolves well in alcohol because iodine and alcohol molecules attract each other.
- Salt dissolves in water because water molecules strongly attract Na^+ and Cl^- ions.



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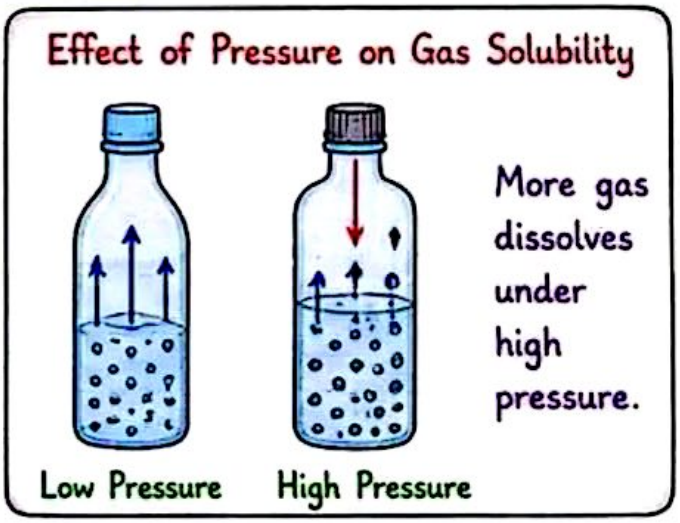
22. Effect of Pressure on Solubility:

Pressure has a significant effect only on gaseous solubility.

- Increase in pressure increases the solubility of a gas in a liquid.
- Decrease in pressure decreases the solubility of a gas in a liquid.

Example:

More CO_2 gas dissolves in soft drinks when the bottle is sealed under high pressure.



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23. Effect of Temperature on Solubility:

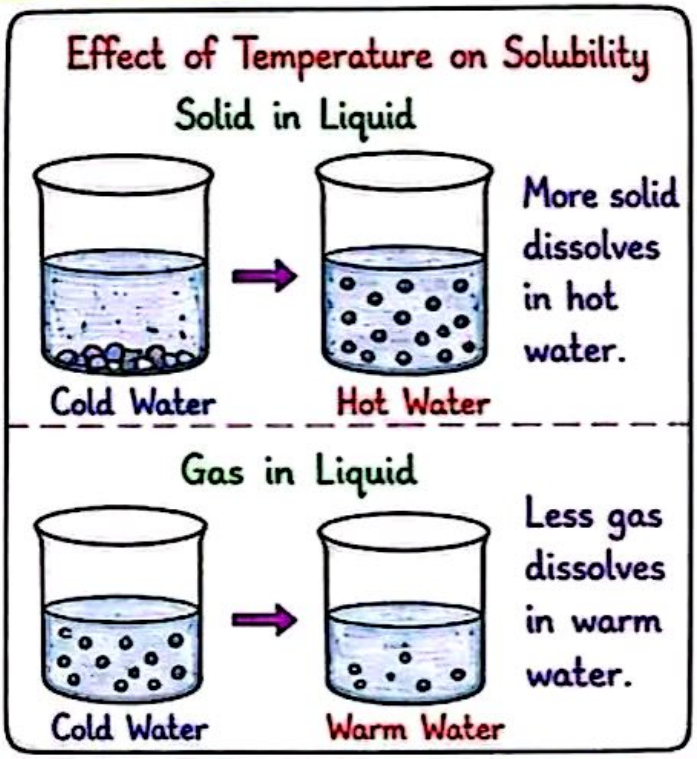
Temperature affects the solubility of solids and gases differently.

- Increase in temperature increases the solubility of most solid substances in liquids.
- Increase in temperature decreases the solubility of gases in liquids.

Example:

Sugar dissolves more in hot water.

Less CO_2 gas dissolves in warm water.



24. Unsaturated, Saturated and Supersaturated Solution:

★ Unsaturated Solution: Contains less solute than the maximum amount that can be dissolved.

★ Saturated Solution: Contains the maximum amount of solute that can be dissolved at a given temperature.

★ Supersaturated Solution: Contains more solute than the maximum amount that can be dissolved at a given temperature.

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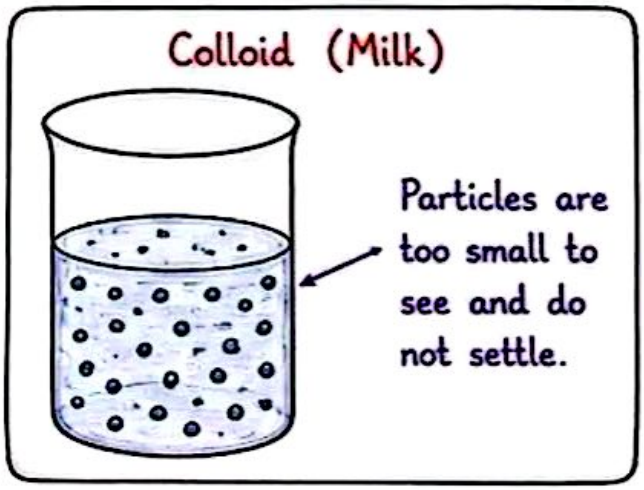


25. Colloids:

A colloid is a type of mixture in which very small particles of one substance are evenly spread in another substance. These particles do not settle down.

Examples:

Milk, Fog, Smoke, Cream, Ink.



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26. Suspension:

A suspension is a heterogeneous mixture in which insoluble particles are large enough to settle down if left undisturbed.

Examples:
Mud in water, Chalk in water, Flour in water.

Suspension (Mud in Water)



Particles settle at the bottom on standing.

27. Difference Between Solution, Colloid and Suspension:

Property	Solution	Colloid	Suspension
Nature	Homogeneous	Heterogeneous	Heterogeneous
Particle Size	Less than 1 nm	1 nm to 1000 nm	More than 1000 nm
Visibility	Particles are not visible	Particles are not visible to naked eye	Particles are visible to naked eye
Stability	Stable, do not settle down	Fairly stable, do not settle down	Unstable, particles settle down
Filtration	Pass through filter paper	Pass through filter paper	Do not pass through filter paper
Example	Salt water, Sugar solution	Milk, Fog, Smoke	Mud in water, Chalk in water

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28. Homogeneous and Heterogeneous Mixture:

Homogeneous Mixture:
A mixture that has the same composition throughout.
Example:
Salt solution, Air.

Homogeneous (Salt Solution)



Same throughout.

Heterogeneous Mixture:
A mixture that has different composition in different parts.
Example:
Oil and water, Sand and water.

Heterogeneous (Oil and Water)



Different in different parts.

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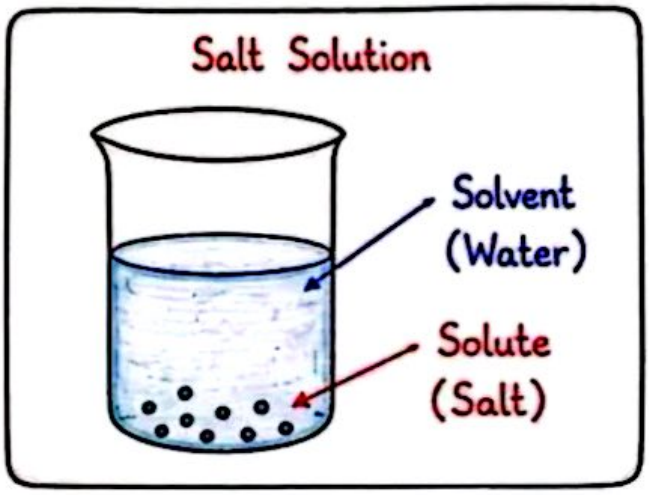
29. Solute and Solvent:

In a solution, the substance that gets dissolved is called the solute and the substance in which it gets dissolved is called the solvent.

Example:

In salt solution:

- Salt is the solute.
- Water is the solvent.



30. Solution, Suspension and Colloid:

Solution	Suspension	Colloid
• Homogeneous mixture. • Particles are very small (less than 1 nm). • Particles do not settle down. • Particles are not visible to naked eye. • Do not scatter light. • Example: Salt solution, Sugar solution.	• Heterogeneous mixture. • Particles are large (more than 1000 nm). • Particles settle down on standing. • Particles are visible to naked eye. • Scatter light. • Example: Mud in water, Chalk in water.	• Heterogeneous mixture. • Particles are intermediate in size (1 nm to 1000 nm). • Particles do not settle down. • Particles are not visible to naked eye. • Scatter light. • Example: Milk, Fog, Smoke.

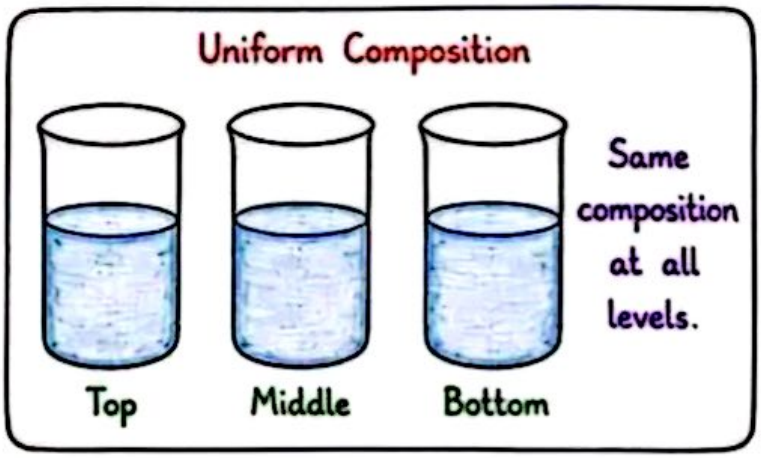
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31. Nature of Solution:

Solutions are homogeneous mixtures. They have uniform composition throughout.

Characteristics:

- Have same composition throughout.
- Particles are very small.
- Do not settle down.
- Do not scatter light.

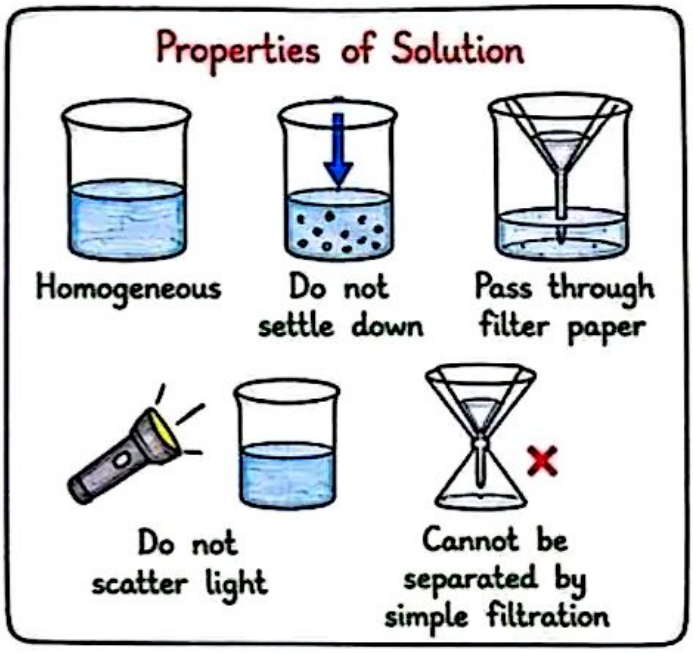


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32. Properties of Solutions:

Some important properties of solutions are:

- They are homogeneous.
- They do not settle down.
- They pass through filter paper.
- They do not scatter light.
- They cannot be separated by simple filtration.

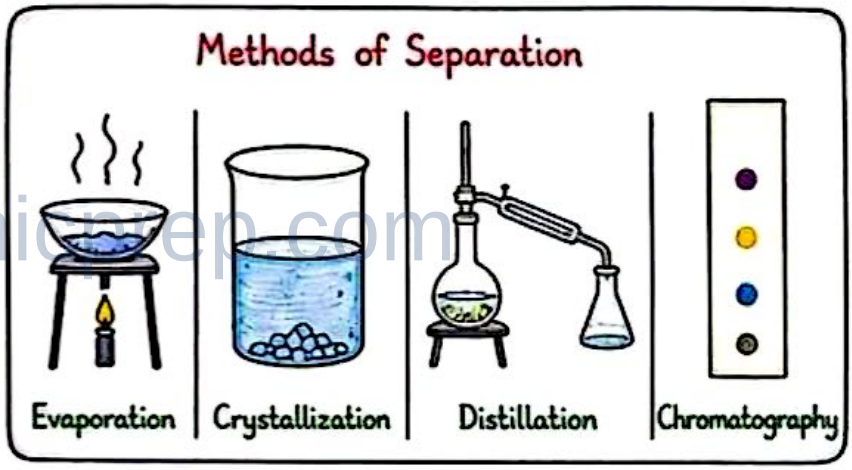


33. Methods of Separation of Solutions:

Solutions cannot be separated by simple filtration. They can be separated by the following methods:

1. Evaporation
2. Crystallization
3. Distillation
4. Chromatography

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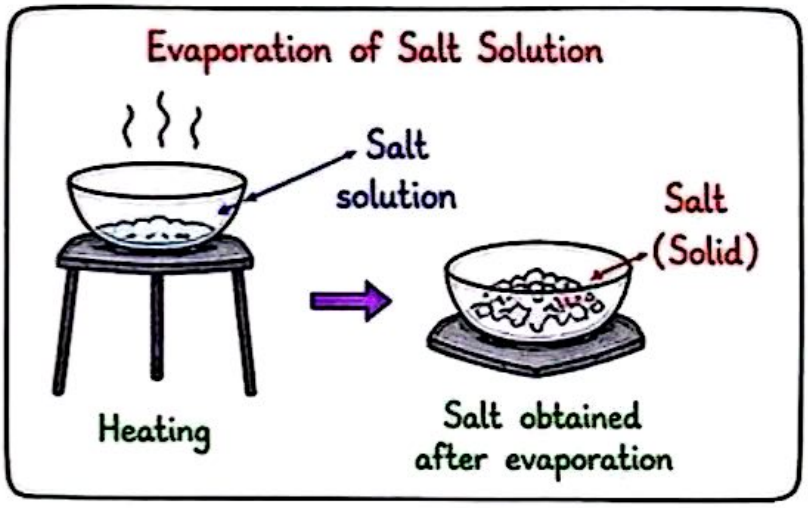


34. Evaporation:

The method of separating a soluble solid from its solution by heating the solution so that the solvent evaporates, leaving the solid behind.

Example:

Salt can be obtained from salt solution by evaporation.



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